Legal, Regulatory & Accessibility Requirements – LIMO Platoon (Market Launch)

Oct 31, 2026 · @Oscar

Going to market changes the legal picture completely. As the **manufacturer**, we must CE-mark the robots under the Machinery Regulation (from 20 January 2027) and the Radio Equipment Directive, and later the Cyber Resilience Act. The AI that drives them becomes a **high-risk AI system** under the AI Act. The research exemptions used for the lab version no longer apply. The RGPD stays central, but our customers become the data controllers, and we are their **processor** for the hosted dashboard and recordings.

1. Product and market assumptions

- **Product:** a fleet of autonomous platooning robots, industrialised from the AgileX LIMO PRO prototype (NVIDIA Orin Nano, LiDAR, depth camera, IMU, ROS 2 on Ubuntu). It is sold with the *Platoon Control Center*, a supervision and teleoperation dashboard offered as a subscription (SaaS) and hosted in an EU cloud region.
- **Modes:** autonomous by default, following a mapped route and traffic rules. Teleoperated by a remote driver when needed. Emergency failsafe for hard braking and platoon splits, where the rear group stops until an operator takes over.
- **Customers:** businesses only (B2B), such as logistics yards, ports, industrial parks and campuses. The robots run on **private sites with restricted access**. Operators are the customers' employees.
- **Market:** Portugal first, then the rest of the EU. The company is a small Portuguese company, possibly a UA spin-off.
- **Out of scope:** driving on public roads, and sales to consumers. If either changes, sections 6.1, 6.6 and 7 need revision.
- **Dashboard:** the camera image fills the screen and telemetry sits on a HUD overlay. The robots stop safely on their own if the cloud or the network fails.

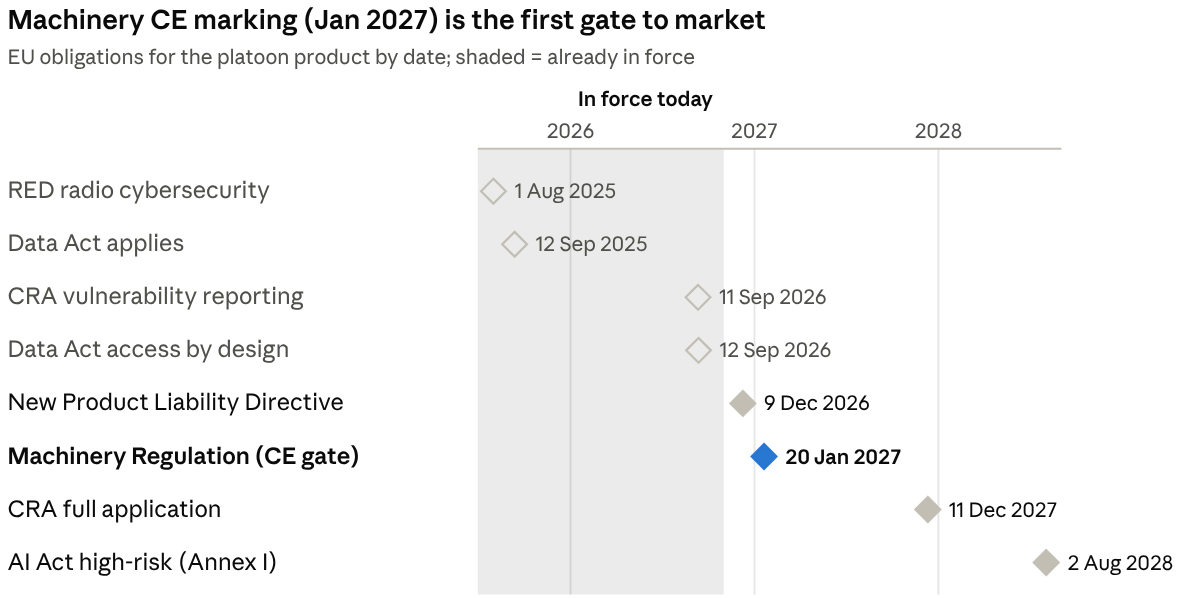
2. Compliance at a glance

Nearly every rule that was "not applicable" to the lab prototype applies to a product on the market. The first hard gate is CE marking under the Machinery Regulation, which needs a notified body because the robots' safety functions use machine learning.

Rule	Our role	Applies from	Main obligation for us
RGPD (EU) 2016/679 + Lei 58/2019	Processor for customers' video and logs; controller for our own accounts and telemetry	In force	Data processing agreements, privacy by design, help customers with DPIAs and data-subject rights (section 3)
Data Act (EU) 2023/2854	Manufacturer / data holder of a connected product; cloud service provider	12 Sep 2025; access by design for products placed after 12 Sep 2026	Customers get access to the data their robots generate; cloud switching made easy (section 6.7)
AI Act (EU) 2024/1689	Provider of a high-risk AI system (safety component of machinery)	2 Aug 2028 for Annex I products (Digital Omnibus)	Risk management, data governance, logging, human oversight, conformity assessment (section 6.2)
Machinery Regulation (EU) 2023/1230	Manufacturer	20 Jan 2027	Essential safety requirements; EU type-examination by a notified body for ML-based safety functions; CE marking (section 6.3)

Rule	Our role	Applies from	Main obligation for us
Radio Equipment Directive 2014/53/EU + Delegated Reg. 2022/30	Manufacturer of radio equipment	In force (cybersecurity since 1 Aug 2025)	CE marking, EN 18031-1 and -2 (section 6.4)
Cyber Resilience Act (EU) 2024/2847	Manufacturer of a product with digital elements	Reporting since 11 Sep 2026; full from 11 Dec 2027	Secure by design, vulnerability reporting, security updates for at least 5 years, SBOM (section 6.5)
Product Liability Directive (EU) 2024/2853	Manufacturer of hardware and software	Products placed on the market after 9 Dec 2026	Strict liability, including for software and updates; insurance (section 6.6)
NIS2 / DL 125/2025	Supplier to essential entities (ports, transport); possibly in scope ourselves as we grow	In force	Supply-chain security clauses from customers (section 6.5)
Código da Estrada / ZLT, DL 67/2021	Not applicable on restricted private sites; ZLT for pilots in real environments	In force	Sell only for sites closed to public traffic (section 6.1)
Software licences	Distributor of software on the robots and the SaaS	In force	Copyleft obligations now trigger; no AGPL code (section 5)
DL 446/85 (standard contract terms)	Supplier using standard B2B terms	In force	Communicate and explain the terms; no exclusion of liability for intent or gross negligence (section 7)

Rule	Our role	Applies from	Main obligation for us
Accessibility: WCAG 2.2 AA / EN 301 549	Software vendor to employers and public buyers	Public procurement and employer duties	Not required by the EAA for B2B tools, but demanded in practice (section 8)



EU compliance dates for the product · 2025–2028

Four obligations already apply (radio cybersecurity, the Data Act, CRA reporting). Product liability and machinery CE marking follow within three months, so a 2027 launch must start the notified-body process now.

3. Data protection (RGPD / GDPR)

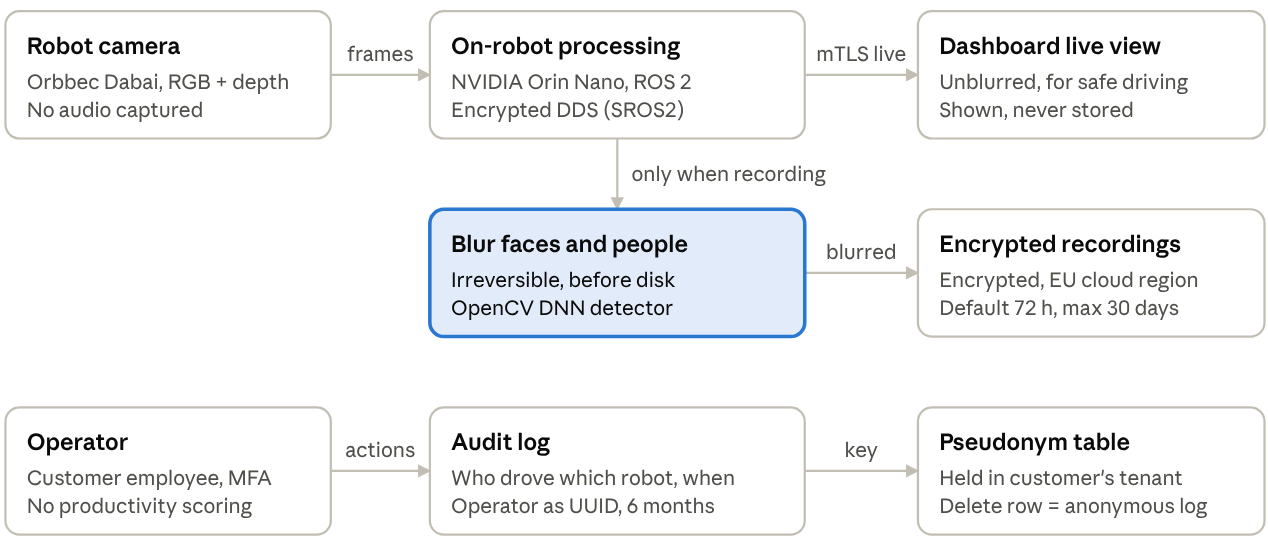
The RGPD applies in full. The robots film people on customer sites, and the dashboard records which employee drove which robot. The roles split as follows:

- **The customer is the controller.** It decides where the robots drive and why they film.
- **We are its processor.** We host the dashboard, recordings and logs (art. 28).
- **We are a controller in our own right** for customer accounts, billing, support and product-improvement data.

In Portugal the RGPD is complemented by Lei 58/2019, and the supervisory authority is the CNPD.

3.1 Roles, contracts and legal basis

- **Data Processing Agreement (art. 28(3)):** part of every customer contract. It says we act only on the customer's instructions and keep staff under confidentiality. It also lists our sub-processors (EU cloud hosting, support tools), commits us to help with DPIAs and data-subject requests, and requires us to delete or return the data at the end of the contract.
- **Customer's legal basis for filming:** usually legitimate interest in safe operation (art. 6(1)(f)), backed by a balancing test that we supply as a template. A public-sector customer, such as a port authority or a university, relies instead on its public task (art. 6(1)(e)).
- **Operators are employees:** the Código do Trabalho forbids remote surveillance to control work performance. Lei 58/2019 (art. 28) restricts the use of recorded images in disciplinary matters. The product therefore offers **no productivity metrics per operator**, and logs serve safety and accountability only. Customers must inform workers and their representatives.
- **Our own processing as controller:**
  - Accounts and billing: performance of the contract (art. 6(1)(b)).
  - Product-improvement telemetry: legitimate interest, pseudonymised.
  - Training our AI models on customer footage: **only if the customer opts in**, only blurred or anonymised data, and only under a separate agreement.
- **DPIA:** customers deploying in areas open to visitors are probably doing systematic monitoring of a publicly accessible area (art. 35(3)(c)), so they need a DPIA. We ship a pre-filled DPIA covering the product.
- **Our DPO:** hosting video for many customers can become large-scale, systematic monitoring (art. 37(1)(b)). We designate a Data Protection Officer before launch.
- **Privacy by design and default (art. 25):** blurring is on, recording is off, and retention is set to the shortest period by default. The customer has to actively change any of these.



personal-data flow · camera and operator data

Raw video goes only to the operator's screen. Anything uploaded to our EU cloud is blurred first, and operator identities stay in the customer's tenant, apart from the logs we can see.

3.2 What personal data the product processes, and why

Data	Source	Personal?	Purpose	Role	Minimisation
Live RGB/depth video	Robot camera	Yes (bystanders, workers)	Real-time supervision and teleoperation	Processor	Relayed encrypted, never stored. No audio
Recordings (rosbag/video clips)	Camera + robot topics	Yes, until blurred	Incident analysis, customer debugging	Processor	Blurred on the robot before upload; recording off by default
Incident clips (collision, hard brake, e-stop)	Automatic trigger	Yes, blurred	Safety investigation, liability evidence	Processor	Short window around the event only (e.g. −30 s / +10 s)
LiDAR point clouds	LiDAR	Practically no	Obstacle detection	—	No texture or colour

Data	Source	Personal?	Purpose	Role	Minimisation
Telemetry, V2V and AI event logs	Robot	No, unless linked to an operator	Control, AI Act logging, Data Act access	Processor	Robot ID, not person ID
Operator accounts (name, work e-mail, role, MFA)	Customer admin	Yes	Access control, training records	Processor	Only work data; no private contacts
Control and audit log	Dashboard	Yes (pseudonymised)	Human-oversight traceability, incident accountability	Processor	Operator UUID; identity table held in the customer's tenant
Connection and security logs (IP, device)	Cloud	Yes	Security monitoring, CRA incident handling	Processor	Short retention
Customer contacts, billing, support tickets	Sales and support	Yes	Contract, invoicing	Controller	Business contacts only
Product-improvement telemetry	Robots (opt-out possible)	Pseudonymised	Reliability, defect detection	Controller	No images; aggregated after 24 months

We do **not** collect audio, biometric templates or face recognition, and we do not track people's locations. The product never scores or profiles individual workers.

3.3 Anonymisation and pseudonymisation

- **On-robot blurring before any upload:** frames are blurred by a person/face detector on the Orin Nano before they leave the robot. The EDPB treats irreversibly blurred footage as erased personal data (EDPB 3/2019, para 103). The live stream stays unblurred so the operator can see pedestrians, and it is never stored.
- **Pseudonymous operators:** logs carry a random UUID. The UUID-to-identity table lives in the customer's own tenant, so our engineers analysing logs cannot see who drove.
- **AI training data:** only blurred frames from customers who opted in. Metadata is stripped, and dataset documentation is kept as the AI Act requires (art. 10).

3.4 Data-subject rights and erasure

- Requests go to the **customer**, as controller. We forward any request we receive within 5 working days.
- **Self-service tools:** in the dashboard the customer can search recordings by robot, date and time; export a clip with other people blurred (EDPB para 94); delete an operator and their pseudonym mapping; and delete footage on request.
- The legal deadline is one month (art. 12(3)). Incident clips under legal hold are an exception, if documented (art. 17(3)(e)).

3.5 Privacy notices and third parties

- **Customer-facing kit:** a layered sign template for the site ("Autonomous robots with cameras"), with the controller, purpose, contact and a QR code to the full notice. It follows EDPB section 7.
- **Our privacy policy:** covers account holders, billing contacts and product telemetry. It is linked from the login page and the website.
- **Sub-processors:** a public list. Customers get 30 days' notice of changes and can object.
- **Hosting and transfers:** data stays in an EU cloud region. If a sub-processor is US-based, we rely on the EU-US Data Privacy Framework or standard contractual clauses plus a transfer impact assessment.
- **No sale or advertising use** of any data. Requests from authorities are passed to the customer unless the law forbids it.

3.6 Security measures (art. 32)

- **Robot:**
  - Secure boot and signed over-the-air updates.
  - SROS2 / DDS-Security, with authenticated and encrypted topics.
  - Encrypted storage.
  - A safe stop on lost link, independent of the cloud.
- **Network:** WPA3-Enterprise on site. The robots connect outbound only, over TLS 1.3 (mTLS with per-robot certificates).
- **Cloud:** tenant isolation, encryption at rest (AES-256) with managed keys, least-privilege access for staff, and access logging.
- **Dashboard:** SSO or multi-factor authentication for operators and admins, role-based access (viewer, operator, admin, safety manager) and session timeouts.

- **Assurance:** a roadmap to ISO/IEC 27001, annual penetration tests and a public vulnerability disclosure policy (CRA).
- **Breaches:** as processor we notify the customer within 24 hours (art. 33(2) requires "without undue delay"). This lets the customer meet its 72-hour deadline to notify the CNPD.

4. Data retention policy

Video is kept as briefly as possible. Logs and documents are kept as long as product-safety law requires. Customers can shorten any period and lengthen video retention only up to the maximum shown. Every deletion runs automatically and is logged. Where no law sets a period, the rule is "necessary for the purpose" (art. 21 Lei 58/2019). For video, the EDPB's "a few days" guidance applies (EDPB 3/2019, para 121).

Data	Default retention	Allowed range	Reason
Live video stream	0 (memory only)	Not configurable	Needed only in real time
Blurred recordings	72 hours	Up to 30 days	EDPB guidance on video storage
Incident clips (collision, hard brake, e-stop)	Until the incident is closed	Until any claim ends (product liability: 3 years from knowledge)	Safety investigation, legal defence (art. 17(3)(e))
AI system event logs (decisions, mode switches, alerts; no images)	12 months	At least 6 months	AI Act art. 19 (providers) and art. 26 (deployers)
Operator control and audit log (pseudonymised)	6 months	6–12 months	Human-oversight traceability, aligned with the AI logs
Connection and security logs	90 days	30–180 days	Incident detection and CRA/NIS2 reporting
Operator accounts	While active + 30 days	—	Access control

Data	Default retention	Allowed range	Reason
Customer data at end of contract	Returned or deleted within 30 days	—	Art. 28(3)(g) RGPD; deletion certificate issued
Product-improvement telemetry (pseudonymised)	24 months, then aggregated	—	Defect detection, post-market monitoring
Invoices and accounting records	10 years	Legal minimum	Portuguese tax law
Technical file, EU declarations of conformity, AI technical documentation	10 years after the last unit is placed on the market	Legal minimum	Machinery Regulation, RED, CRA, AI Act art. 18 (not personal data)
AI training datasets (blurred and anonymised only)	Life of the model version + 10 years	—	AI Act data-governance evidence; no personal data kept

Retention is enforced by scheduled purge jobs in the cloud and on each robot. The purge logs are part of our RGPD accountability file (art. 5(2)).

5. Licensing and intellectual property

Selling robots with software installed is **distributing** that software, so the copyleft conditions that never triggered in the lab now apply. Our own code can stay proprietary because the core stack is Apache-2.0 or BSD. Three items need action before launch: no AGPL code (Ultralytics YOLO), GPL source offers for the robot's Linux image, and a commercial agreement with AgileX and Canonical for the base platform.

Component	Licence	Commercial impact	Action
ROS 2 core, Fast DDS	Apache-2.0	Allowed, with patent grant	Ship licence texts and NOTICE files in the product documentation and the "About" screen

Component	Licence	Commercial impact	Action
Gazebo	Apache-2.0	Development tool, not shipped	None
Nav2	Mostly Apache-2.0; AMCL LGPL-2.1	LGPL parts allowed if dynamically linked	Publish any changes to LGPL parts; allow relinking
Eclipse Cyclone DDS (if used)	EPL-2.0	Allowed	Publish changes to Cyclone itself
orbbec_camera driver	Apache-2.0	Allowed	Attribution
LIMO ROS 2 packages, EAI/YDLIDAR driver	Not yet confirmed	Unknown	Check LICENSE files; sign an OEM / distribution agreement with AgileX
OpenCV 4.5+	Apache-2.0	Allowed	Attribution
Ultralytics YOLO	AGPL-3.0 or paid Enterprise licence	AGPL would force us to publish our whole robot and SaaS source, because its network clause covers SaaS	Do not ship. Use YOLOX or RT-DETR (Apache-2.0), or buy the Enterprise licence
Pretrained weights and datasets (e.g. COCO)	Various (CC-BY, research-only)	Some forbid commercial use	Train on our own or commercially licensed data; keep a dataset licence register
NVIDIA JetPack, CUDA, TensorRT	NVIDIA proprietary licence	Runtime redistribution on Jetson-based products allowed under its terms	Review the licence terms; no modification of the binaries

Component	Licence	Commercial impact	Action
Ubuntu, Linux kernel, BusyBox, etc.	GPL-2.0, LGPL and others	Shipping a device with Linux counts as distribution	Written offer of source for GPL components, plus a build manifest. Canonical's IP policy restricts shipping modified images under the Ubuntu name: use Ubuntu Pro for Devices or an unbranded or Yocto image
Dashboard: React, roslibjs, rosbridge_suite	MIT / BSD	Allowed	"Open-source licences" page in the web app
Cloud components (databases, caches)	Check each one	SSPL / BSL / AGPL licences can restrict SaaS use	Choose permissively licensed services, or buy commercial licences

Other IP actions:

- **Ownership of student code:** code written during the course belongs to the authors or UA under UA's IP rules. It must be assigned or licensed to the company before launch.
- **Patents:** truck platooning and teleoperation are heavily patented. Run a freedom-to-operate search (Espacenet, EPO) before launch. Consider filing for our split-and-takeover protocol at INPI or the EPO **before** any public demo or thesis, because public disclosure destroys novelty.
- **Trademarks:** register the product name at INPI, then the EUIPO. "LIMO", "AgileX" and "NVIDIA" are used only to describe the hardware, as the OEM agreement allows.
- **Our licence model:** a proprietary EULA for the robot software, SaaS terms for the dashboard, and an optional open-source SDK under Apache-2.0.
- **Release checks:** at every release we generate the third-party licence report and the **SBOM**, which the CRA also requires, from the same scan (e.g. `scancode`, `syft`).

6. Sector-specific regulation (autonomous mobile machinery)

As a product on the EU market, the platoon is regulated as **machinery with high-risk AI and radio**. It is also a **connected product** under the Data Act and a **product with digital elements** under the CRA. Each regime has its own conformity duties, but they share one technical file and one CE marking.

6.1 Where the robots may drive

- **Código da Estrada:** it covers public roads *and private roads open to public traffic*. Portugal has no legal regime for autonomous vehicles (AMT Opinion 92/2023). So we sell only for sites **closed to public traffic** (fenced yards, port terminals, industrial areas). The contract and the instructions state this.
- **Pilots in real environments:** these go through a Zona Livre Tecnológica (DL 67/2021). The Aveiro ZLT includes the UA campus, the Port of Aveiro and a section of the A25. It needs authorisation from the zone manager, insurance and a public-information plan.
- **Public roads (long term):** these would require EU vehicle type-approval (Reg. (EU) 2018/858 and the automated-driving rules of Reg. (EU) 2022/1426) and compulsory motor insurance. This is outside the business plan.

6.2 Artificial intelligence (AI Act)

- **Classification:** the perception and control AI is a safety component of machinery that needs third-party conformity assessment. That makes it **high-risk** under art. 6(1) and Annex I. We are the **provider**; the customer is the **deployer**.
- **Date:** the 2026 Digital Omnibus moved high-risk obligations for Annex I products to **2 August 2028** (CASRAI). We build compliance in now, because the Machinery Regulation already asks for much of it in 2027.
- **Provider obligations:**
  - Risk management (art. 9) and data governance of training sets (art. 10).
  - Technical documentation (art. 11) and automatic logging (art. 12).
  - Instructions for use (art. 13) and human oversight (art. 14).
  - Accuracy, robustness and cybersecurity (art. 15), and a quality management system (art. 17).
  - Post-market monitoring (art. 72) and serious-incident reporting (art. 73).
- **The dashboard is our art. 14 measure:**
  - Operators see the robot's state and its confidence.

  - They can override or stop any robot at any moment.
  - The system asks for a human when the platoon splits.
- **Deployer duties we support (art. 26):** trained operators (we provide certified training), use according to the instructions, and keeping logs for at least 6 months.

6.3 Machine safety (Machinery Regulation (EU) 2023/1230)

- **Timing:** from **20 January 2027** we are the manufacturer of autonomous mobile machinery (EU-OSHA). Units placed on the market before then fall under the old Machinery Directive (DL 103/2008).
- **Notified body required:** the safety functions (obstacle stop, following distance, split detection) rely on machine-learning perception. That falls under Annex I Part A, so **EU type-examination by a notified body is mandatory**; self-certification is not allowed (DTI).
- **Essential requirements we design for:**
  - A remote supervisor function for autonomous mobile machinery.
  - An emergency stop that works without the network, and a safe stop on loss of communication.
  - Speed and separation limits near people.
  - Protection against corruption, including cyber attacks on safety functions.
  - Ergonomic operator interfaces.
- **Standards:** ISO 12100 (risk assessment), ISO 13849-1 / IEC 62061 (functional safety of the e-stop and safety controller) and ISO 3691-4 (driverless industrial vehicles).
- **Documentation:** a technical file kept for 10 years, an EU declaration of conformity, CE marking, and instructions in Portuguese for the Portuguese market (digital allowed, with paper on request).

6.4 Radio, electrical and environmental product rules

- **Radio Equipment Directive (DL 57/2017 in Portugal):** we integrate the radio, so we are the manufacturer. That means CE marking and the cybersecurity requirements of Delegated Reg. 2022/30, which apply since 1 August 2025 (SGS): EN 18031-1 for internet connection and EN 18031-2 because the camera processes personal data.
- **Spectrum:** use only bands authorised by ANACOM. For dependable V2V, consider the harmonised 5.9 GHz ITS band (ITS-G5 / C-V2X) instead of shared Wi-Fi. ETSI EN 302 637-2/-3 (CAM/DENM) define the message format.

- **Electrical and environmental rules:**
  - EMC Directive 2014/30/EU and the Low Voltage Directive for the chargers.
  - Batteries Regulation (EU) 2023/1542: labelling, collection, due diligence.
  - WEEE: register as a producer in Portugal and join a collection scheme.

6.5 Cybersecurity

- **Cyber Resilience Act (EU) 2024/2847:** the robot and its remote-control software are a product with digital elements (Finite State).
  - **Since 11 September 2026:** report actively exploited vulnerabilities and severe incidents to ENISA and the national CSIRT (CNCS) within 24 h (early warning), 72 h (notification) and 14 days or one month (final report).
  - **From 11 December 2027:** the essential requirements and CE marking apply. These include security by design, no known exploitable vulnerabilities at release, secure defaults, a support period of at least 5 years with free security updates, an SBOM and a vulnerability disclosure policy.
- **NIS2 / DL 125/2025:** a small company is probably outside it at launch. Manufacturing of machinery is a covered sector once we reach medium size. Our customers (ports, transport, logistics) are covered, so they will impose supply-chain security clauses and audits.

6.6 Liability and insurance

- **New Product Liability Directive (EU) 2024/2853:** applies to products placed on the market after 9 December 2026 (Cuatrecasas). Until Portugal transposes it, DL 383/89 applies.
  - Software, AI and updates count as the product.
  - Not supplying a security update can make the product defective.
  - Defect and causation are presumed in some cases.
  - Claims are possible for 3 years from knowledge, within 10 years of placing on the market (25 for latent personal injury).
  - This liability **cannot be excluded by contract**.
- **Measures:**
  - Product liability and professional indemnity insurance.
  - Incident logs kept as evidence (section 4).
  - A recall and field-update procedure.

- Post-market monitoring shared with the AI Act process.

6.7 Data access (Data Act)

- The robots are **connected products** and the Control Center is a **related service** (WSGR):
  - **Before the contract:** tell customers what data the robots generate and how to access it.
  - **On request:** give users free access to product data (telemetry, sensor data, logs) in a machine-readable format, and share it with a third party they choose. Trade-secret safeguards apply.
  - **Products placed on the market after 12 September 2026:** access by design.
  - **Cloud switching rules for the SaaS:** export tools, no lock-in, and switching charges phased out.
- **Implementation:** a documented data API and a "Download my fleet data" export in the dashboard.

7. Terms and conditions (key points)

The contract set is a **Master Agreement** for the robots, **SaaS terms** for the Control Center, a **EULA** for the robot software, the **Data Processing Agreement** and an **SLA**. These are standard terms used with many customers, so DL 446/85 (*cláusulas contratuais gerais*) applies. We must communicate them in full and explain them before signature, and some clauses are void by law.

- **Permitted use:**
  - Only on sites closed to public traffic, after the customer has done a site risk assessment.
  - Only by operators trained and certified by us.
  - Within the configured speed and area limits.
- **Prohibited use:**
  - Public roads.
  - Disabling or bypassing safety functions or the e-stop.
  - Modifying firmware or reverse engineering.
  - Sharing credentials.
  - Using the cameras to monitor workers' performance.

    - Re-export in breach of sanctions.
- **Customer obligations:**
    - Site signage and the privacy notice; a DPIA where needed.
    - Informing workers and their representatives.
    - Insurance for site operations.
    - Installing safety and security updates within the stated time.
- **Warranty:** 24 months on hardware defects. Security updates during the declared support period, at least 5 years under the CRA. Beta features are offered "as is", clearly labelled.
- **Liability:**
    - Contractual liability is capped (e.g. at the fees paid in the last 12 months) and indirect losses are excluded.
    - By law we **cannot** exclude liability for intent or gross negligence, or for death and personal injury (DL 446/85, art. 18; Código Civil, art. 809).
    - Product liability under the PLD cannot be limited by contract.
- **SLA:** a dashboard availability target (e.g. 99.5% a month) with service credits. Safety never depends on the cloud, because the robots stop safely on their own.
- **Payment, refunds and cancellation:**
    - The SaaS is billed monthly or yearly and can be cancelled with 30 days' notice.
    - If we materially breach the contract, prepaid annual fees are refunded pro rata.
    - Hardware is sold or leased under the Master Agreement.
    - B2B buyers have no legal right of withdrawal. If we ever sell to consumers, the 14-day withdrawal right (DL 24/2014) and the 3-year legal guarantee (DL 84/2021) would apply.
- **Data:** the DPA governs personal data. Customers keep the rights to their data under the Data Act, with export at any time and return or deletion within 30 days of termination.
- **IP:** we keep all rights in the software and AI models; the customer gets a licence to use them for the subscription term.
- **Suspension:** we may remotely restrict a robot or the service when there is an imminent safety or security risk, or when payment is overdue after notice.
- **Law and jurisdiction:** Portuguese law, with the CISG excluded for the hardware sales. Disputes go to the courts of Aveiro, or to commercial arbitration if both parties agree.

- **Changes to the terms:** 30 days' notice of material changes, with the right to terminate without penalty if the customer disagrees. Safety-critical changes, such as new usage restrictions after an incident, take effect immediately. The version and date are shown in the dashboard.

8. Usability and digital accessibility

Accessibility is now a **sales requirement**, not only a course requirement. No accessibility law directly covers a B2B supervision tool. However, public buyers must demand accessibility in tenders, and employers must accommodate operators with disabilities. Our target is **WCAG 2.2 level AA**, and **EN 301 549** for any native client, documented in a public conformance report.

The main weakness remains that all information sits on top of a moving video. Contrast changes every frame, status is shown by colour, and alerts are visual or audio-only. Driving from a camera image is inherently visual, but supervision, alerts, mode switching, the emergency stop, fleet management and logs can all be made fully accessible.

8.1 Legal and market framing

- **European Accessibility Act / DL 82/2022** (since 28 June 2025): it covers listed consumer products and services such as banking, e-commerce and passenger transport. Industrial supervision software is not on the list, so it is not mandatory. It *would* apply to consumer-facing parts if we ever offered robot shuttles for passengers, such as apps, e-tickets or real-time information.
- **Public procurement:** ports, municipalities and universities must include accessibility in their technical specifications (Directive 2014/24/EU, transposed in the Código dos Contratos Públicos). In practice they ask for EN 301 549 conformance, so without it we lose tenders.
- **DL 83/2018:** it applies to public bodies' own websites and apps. A public customer that offers our dashboard to its staff or the public will require the same WCAG AA level and an accessibility statement.
- **Employment law:**
  - The operators are our customers' employees. Employers must make reasonable accommodation for workers with disabilities (Directive 2000/78/EC; Código do Trabalho).
  - Lei 4/2019 sets employment quotas for people with disabilities in companies with 75 or more workers.

    - An accessible supervision role is a selling point.
- **Machinery Regulation:** it requires ergonomic operator interfaces that reduce fatigue and stress, which overlaps with WCAG for the HUD.

8.2 User profiles of the system

System role	Main task	Context
Remote supervisor / teleoperator (customer employee)	Supervise several platoons, take over the leader or a split group, press e-stop	Shift work and time-critical events; keyboard, mouse or gamepad; long sessions
Fleet admin (customer)	Configure sites, routes and users; review logs; export data	Not time-critical; forms and tables
Site safety manager (customer)	Review incidents, set speed and area limits, approve operators	Reports and incident clips
Our support engineer	Diagnose robots remotely, with the customer's permission	Pseudonymised logs only
Visitor / observer	Watch a demo or a public display	Read-only; far from the screen

8.3 Screens analysed

1. **Login**: username, password and terms acceptance.
2. **Main view**: full-screen camera with HUD overlay (speed, mode, battery, distance to vehicle ahead, V2V link quality, platoon map strip) and teleoperation controls.
3. **Platoon / robot selector**: list of robots with status, used to switch the camera or take control.
4. **Alert and takeover dialog**: hard brake, platoon split and "operator needed" requests, with a countdown.
5. **Logs and settings (admin)**: users, recordings, retention and HUD preferences.

8.4 Barriers by profile

Profile	Main barrier in our dashboard	Question for each screen	Our answer
Blind	Driving depends on video; icon-only HUD; alerts not announced	Can I use everything with a screen reader?	No for driving (inherently visual). Yes for supervision: alerts in `aria-live`, a text status panel, named buttons
Low vision	Small HUD text over video; layout breaks at 200% zoom	Still readable at 200% zoom?	HUD scale setting; text on an opaque backplate; reflow at 200%
Colour blindness	Green/amber/red link and robot states; red-only alerts	Can I understand everything without colour?	Icon + text label + shape for every state
Motor impairment	Drag-only virtual joystick; small buttons; hold-to-drive	Can I do it with the keyboard only?	Full keyboard control, remappable keys, 44×44 px targets
Deaf / hard of hearing	Hard-brake alert is a beep only	Is there a text alternative?	Every sound has a visual banner and screen flash below 3 Hz
Cognitive / low digital literacy	Jargon ("RSSI", "V2V", "split"); unclear takeover flow	Clear on the first try?	Plain labels, a one-action takeover, consistent terms, help tooltips

8.5 Accessibility matrix (screen · element)

Screen · element	Profile affected	Problem	WCAG 2.2 criterion	Severity	Proposed fix
Main view · HUD text over video	Low vision, all users	White text on a bright or changing background drops below 4.5:1	1.4.3 Contrast (Minimum)	High	Semi-opaque dark backplate (at least 80%) behind every HUD element; text white, 4.5:1 or more guaranteed
Main view · link and robot status dots	Colour blind	State shown only by green/amber/red	1.4.1 Use of Color	High	Icon + word ("Link OK", "Weak", "Lost") + shape
Alert dialog · hard brake / split	Blind, all users	Alert appears visually but is not announced	4.1.3 Status Messages	High	`role="alertdialog"` and `aria-live="assertive"`; focus moves to the main action
Alert dialog · audio beep	Deaf	Emergency signalled only by sound	1.1.1 Non-text Content (text alternative)	High	Visual banner + text; optional vibration on gamepad
Main view · emergency stop	Motor, all users	Small icon button, mouse only	2.1.1 Keyboard; 2.5.8 Target Size (Minimum); 4.1.2 Name, Role, Value	High	Large labelled "STOP ALL" button (at least 44×44 px), always visible, keyboard shortcut, `aria-label`
Main view · virtual joystick	Motor	Steering only by dragging	2.5.7 Dragging Movements	Medium	Arrow/WASD keys and on-screen buttons as single-pointer alternatives
Main view · single-key shortcuts (W, A, S, D, Space)	Speech-input and keyboard users	Single keys trigger movement by accident	2.1.4 Character Key Shortcuts	High	Shortcuts active only when "drive mode" is armed; remappable or disable-able in settings
Main view · HUD layout	Low vision	At 200% zoom the HUD covers the video or gets cut off	1.4.4 Resize Text; 1.4.10 Reflow	Medium	Collapsible HUD; HUD scale in settings; side-panel layout at high zoom

Screen · element	Profile affected	Problem	WCAG 2.2 criterion	Severity	Proposed fix
Main view · keyboard focus	Keyboard users	Focus ring invisible on video or hidden under the HUD	2.4.7 Focus Visible; 2.4.11 Focus Not Obscured (Minimum)	Medium	3 px high-contrast focus outline with a dark halo; overlay never covers the focused element
Alert dialog · takeover countdown	Cognitive, motor	10 s limit to accept takeover	2.2.1 Timing Adjustable (real-time exception)	Medium	Countdown in text ("Platoon B stopped — take control? 10 s"); if time runs out the robots stay safely stopped, nothing is lost
Alert dialog · flashing warning	Photosensitive	Red flashing border on hard brake	2.3.1 Three Flashes or Below Threshold	High	At most 2 flashes per second, or a steady colour and icon
Main view · camera feed	Blind	Video has no text equivalent	1.1.1 Non-text Content	Low (inherently visual)	`aria-label` "Live camera, robot Leader"; text status panel with detected obstacles and distances
Selector · robot list	Blind	Rows are clickable `div`s with no role or name	4.1.2 Name, Role, Value; 1.3.1 Info and Relationships	Medium	Native `<button>` or table with headers; name + state in text
Login · fields	Blind, cognitive	Placeholder used as label; error shown in red only	3.3.2 Labels or Instructions; 3.3.1 Error Identification	High	Visible `<label>`; error text next to the field + `aria-describedby`
Login · password	Cognitive	Paste blocked; no password-manager support	3.3.8 Accessible Authentication (Minimum)	Medium	Allow paste and autofill; "show password" button with a name
All screens · structure	Blind	No headings or landmarks; the HUD is read in a random order	1.3.1 Info and Relationships; 2.4.6 Headings and Labels	Medium	`<main>`, `<nav>`, `<h1>` – `<h2>`; logical DOM order: alerts → status → controls
All screens · language	Cognitive	Technical jargon (RSSI, V2V, CACC)	3.1 Readable (good practice)	Low	Plain labels ("Signal", "Vehicle link") with tooltips; consistent terms across screens

8.6 For the presentation: the five most serious issues

1. **HUD contrast over video** (1.4.3): show a frame where the speed text disappears, then the version with a backplate.
2. **Status only by colour** (1.4.1): show the link dots before and after with a colour-blindness simulation (Chrome DevTools › Rendering).
3. **Unannounced emergency alerts** (4.1.3 + 1.1.1): demo with NVDA that "Hard brake — platoon stopped" is read aloud and shown as text.
4. **Emergency stop not keyboard-operable and too small** (2.1.1 / 2.5.8): show the "mouse unplugged" exercise.
5. **Accidental single-key driving** (2.1.4): show the armed drive mode and the remapping setting.

Tools: WebAIM Contrast Checker, Colour Contrast Analyser, axe DevTools, NVDA (Windows) and VoiceOver (macOS). SUS testing comes in a later milestone.

Sources

- Lei 58/2019 — execution of the RGPD in Portugal (Legiscope summary)
- EDPB Guidelines 3/2019 on processing of personal data through video devices (PT)
- Wilson Sonsini — EU Data Act September 2026 deadline
- CASRAI — EU AI Act obligations, exemptions and Digital Omnibus dates
- AI Act Service Desk — Article 19, automatically generated logs
- EU-OSHA — Machinery Regulation (EU) 2023/1230
- DTI — Type examination of machines that use AI for safety functions
- SGS — RED cybersecurity requirements mandatory from 1 August 2025
- Finite State — Cyber Resilience Act timeline 2026–2027
- PwC — DL 125/2025, transposition of NIS2 in Portugal
- Cuatrecasas — New Product Liability Directive 2024/2853
- Mórais Leitão — Decree-Law 67/2021 (Technological Free Zones)
- AMT Opinion 92/2023 — Aveiro ZLT regulation
- Garrigues — DL 82/2022, European Accessibility Act in Portugal
- Accesstive — Decree-Law 83/2018 summary
- Ultralytics licensing

- ROS Index — orbbec_camera package
- Project documents: *Teleoperação de Pelotão de Robots* (proposal), *PEI Milestone 3* brief, FinalArch diagrams.